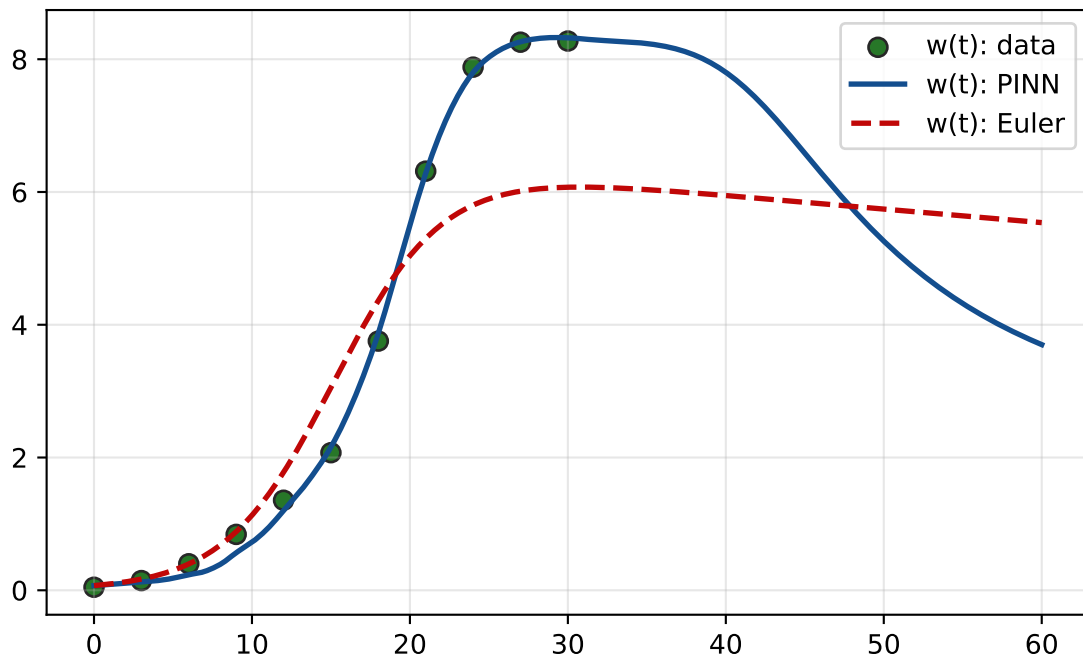
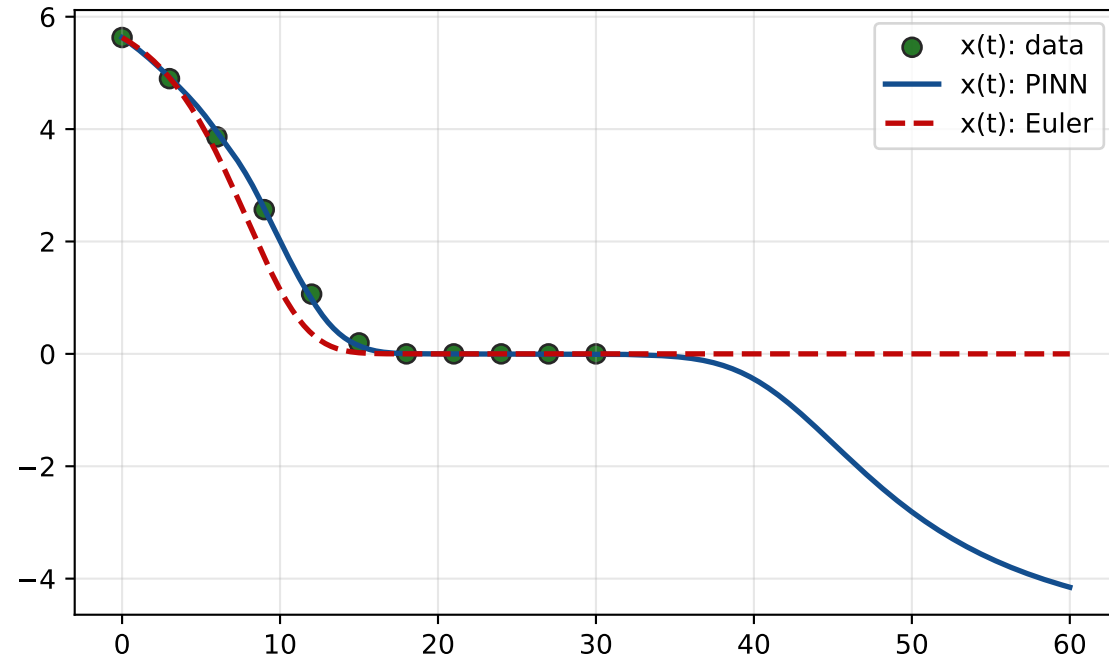


# E6 | PINN vs Euler

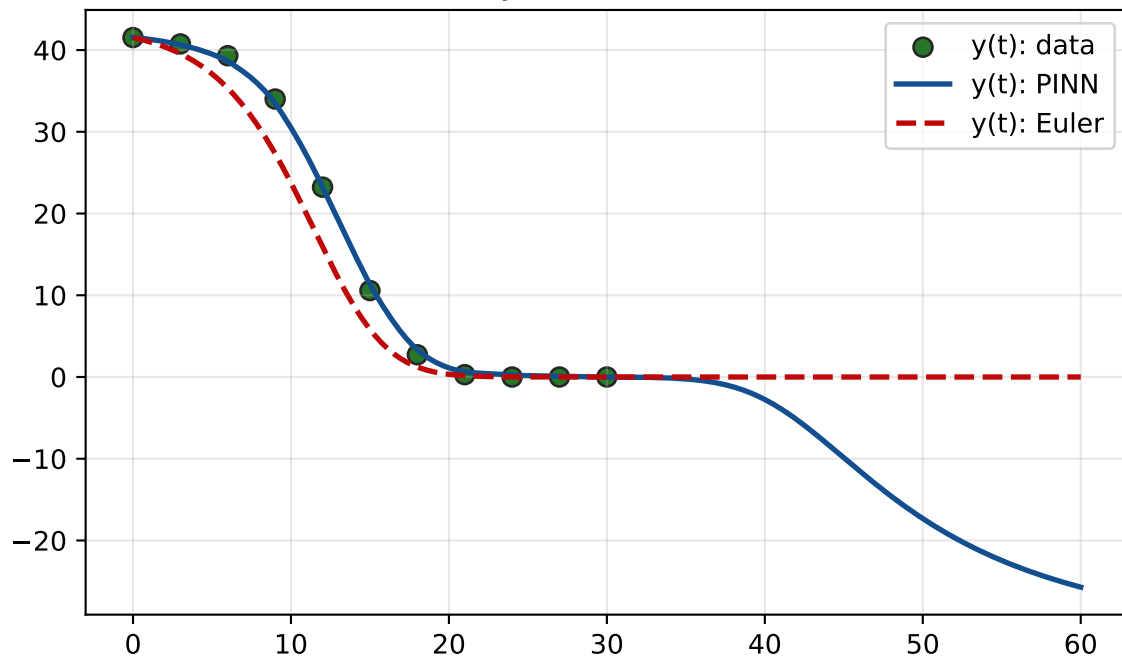
w(t): Biomass



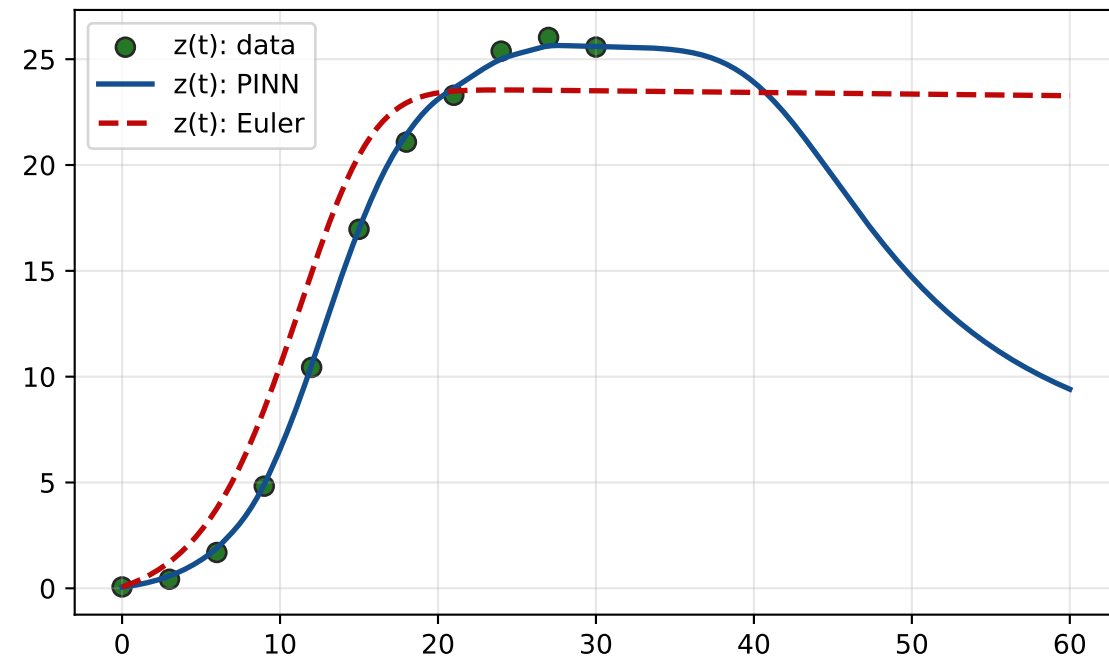
x(t): Glucose



y(t): Fructose



z(t): Ethanol



E6

R2=0.9521, R2\_adj=0.9444, RSS=2.262e+02, AIC=8.632e+01

cond(JTJ)=3.87e+04, used\_pinv=False

| param | estimate     | SE           | MOE_95       | CI_low        | CI_high      | p_value      |
|-------|--------------|--------------|--------------|---------------|--------------|--------------|
| p1    | 2.876250e-01 | 3.063783e-01 | 6.202305e-01 | -3.326055e-01 | 9.078555e-01 | 3.537676e-01 |
| p2    | 4.505631e-02 | 4.266965e-02 | 8.638019e-02 | -4.132388e-02 | 1.314365e-01 | 2.976660e-01 |
| p3    | 1.000000e-02 | 4.142831e-01 | 8.386723e-01 | -8.286723e-01 | 8.486723e-01 | 9.808688e-01 |
| p5    | 4.009600e-01 | 3.926940e-01 | 7.949674e-01 | -3.940073e-01 | 1.195927e+00 | 3.136876e-01 |
| p7    | 1.394657e-01 | 7.757103e-02 | 1.570343e-01 | -1.756866e-02 | 2.965000e-01 | 8.014035e-02 |
| p9    | 7.600548e-02 | 4.178867e-02 | 8.459674e-02 | -8.591255e-03 | 1.606022e-01 | 7.682865e-02 |

| IC             | estimate     |
|----------------|--------------|
| w0 (estimated) | 7.046454e-02 |
| x0 (fixed)     | 5.629963e+00 |
| y0 (fixed)     | 4.152602e+01 |
| z0 (fixed)     | 6.451776e-02 |
| W0 (fixed)     | 0.000000e+00 |